

SHETH L.U.J. & SIR M.V. COLLEGE OF SCIENCE

SUBJECT - Data Analysis with R/SAS/SPSS

Aim - Generating descriptive statistics using summary() or describe() (R).

Output :

The image displays two screenshots of the RStudio interface, showing the execution of R code to generate descriptive statistics for a dataset named 'data'.

Top Screenshot: The console shows the following code and output:

```
> # Clear workspace
> rm(list = ls())
> # Load the dataset
> data <- read.csv("Data.csv")
> # View first few rows
> head(data)
  AT      V      AP      RH      PE
1 14.96 41.76 1024.07 73.17 463.26
2 25.18 62.96 1020.04 59.08 444.37
3  5.11 39.40 1012.16 92.14 488.56
4 20.86 57.32 1010.24 76.64 446.48
5 10.82 37.50 1009.23 96.62 473.90
6 26.27 59.44 1012.23 58.77 443.67
> # Select required columns
> selected_data <- data[, c("AT", "V", "AP", "RH", "PE")]
> # Check data structure
> str(selected_data)
'data.frame':   9568 obs. of  5 variables:
 $ AT: num  14.96 25.18 5.11 20.86 10.82 ...
 $ V : num  41.8 63 39.4 57.3 37.5 ...
 $ AP: num  1024 1020 1012 1010 1009 ...
 $ RH: num  73.2 59.1 92.1 76.6 96.6 ...
 $ PE: num  463 444 489 446 474 ...
> # Descriptive statistics using summary()
> summary(selected_data)
   Min.   1st Qu.  Median    Mean   3rd Qu.    Max.
AT   1.81   13.51   20.34   19.65   25.72   37.11
V    25.36   41.74   52.08   54.31   66.54   81.56
AP  992.9 1009.1 1012.9 1013.3 1017.3 1033.3
RH   25.56   63.33   74.97   73.31   84.83  100.16
PE  420.3  439.8  451.6  454.4  468.4  495.8

> # Install psych package (run only once)
> if (!require("psych")) {
+   install.packages("psych")
+ } else {
+   library(psych)
+ }
```

Bottom Screenshot: The console shows the following code and output:

```
> str(selected_data)
'data.frame':   9568 obs. of  5 variables:
 $ AT: num  14.96 25.18 5.11 20.86 10.82 ...
 $ V : num  41.8 63 39.4 57.3 37.5 ...
 $ AP: num  1024 1020 1012 1010 1009 ...
 $ RH: num  73.2 59.1 92.1 76.6 96.6 ...
 $ PE: num  463 444 489 446 474 ...
> # Descriptive statistics using summary()
> summary(selected_data)
   Min.   1st Qu.  Median    Mean   3rd Qu.    Max.
AT   1.81   13.51   20.34   19.65   25.72   37.11
V    25.36   41.74   52.08   54.31   66.54   81.56
AP  992.9 1009.1 1012.9 1013.3 1017.3 1033.3
RH   25.56   63.33   74.97   73.31   84.83  100.16
PE  420.3  439.8  451.6  454.4  468.4  495.8

> # Install psych package (run only once)
> if (!require("psych")) {
+   install.packages("psych")
+ } else {
+   library(psych)
+ }
```

The Environment pane on the right shows the loaded data objects: 'data' (9568 obs. of 5 variables) and 'selected_data' (9568 obs. of 5 variables).

Output of describe() function:

```
> # Descriptive statistics using describe()
> describe(selected_data)
  vars  n  mean  sd  min  max range skew kurtosis  se
AT    1 9568  19.65  12.25  1.81  37.11 35.30 -0.14  -1.04  0.08
V     2 9568  54.31  12.25  25.36  81.56 56.20  0.20  -1.44  0.13
AP    3 9568 1013.11  59.94 1012.94 1013.11 6.03 992.89 1033.30 40.41 0.27  0.09 0.06
RH    4 9568  73.31  16.00  74.97  74.07 15.65 25.56 100.16 74.60 -0.43 -0.45 0.15
PE    5 9568  451.55  453.64 20.56 420.26 495.76 75.50  0.31  -1.05 0.17
```

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